Ari's Dream Final Project

Project Report

by

CS 174

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Introduction:

I have always wanted to create an instagram like website, and after I read the description of the Ari's Dream project it sounded like a base model for it and made me instantly want to choose this as my final project. Being able to showcase images and your thoughts and it being displayed on a homepage just for you and maybe chosen others in the future is something that sounds really interesting to me. Having a chill vibe where users can share and be themselves is my goal.

Project Overview:

This web application is about expressing yourself easily. You register and log in, and right away are able to add a new post with a handful of images and text and after clicking submit the magic is done in the background and you have your own pesonalized homepage with each post. You can click on each post and see details about each one from every single image you uploaded and text for that post. Currently its setup to be your own personal media space where you share your life with images. If you don't like a post you uploaded, you'll be able to delete them all.

Project Features:

Feature #1: User Registration and Login

- A user-friendly registration which authenticates the user properly taking their name, email, username, password, and option to add a profile picture
- A login system ensuring the security of the user information that checks to make sure the correct user is logging in with the database

Feature #2: Post Creation

- A simple and easy to use interface for users to upload multiple images and text for each of their posts
- Users can see a visual preview of their uploaded images before submission so they can change it if needed
- A dynamic progress bar for visually representing the status of the upload with a percent to keep users informed

Feature #3: Personalized Homepage

- All posts are showcased within the homepage and can be clicked on for a more detailed view to look at every image of each post
- Each user can only view their own posts within the homepage

Feature #4: Detailed Post View

- Upon clicking any post in the homepage, you are taken to another page with a detailed view of the post being able to slide through each image and seeing the text description below
- Each post can be deleted easily if the user wants to remove certain posts

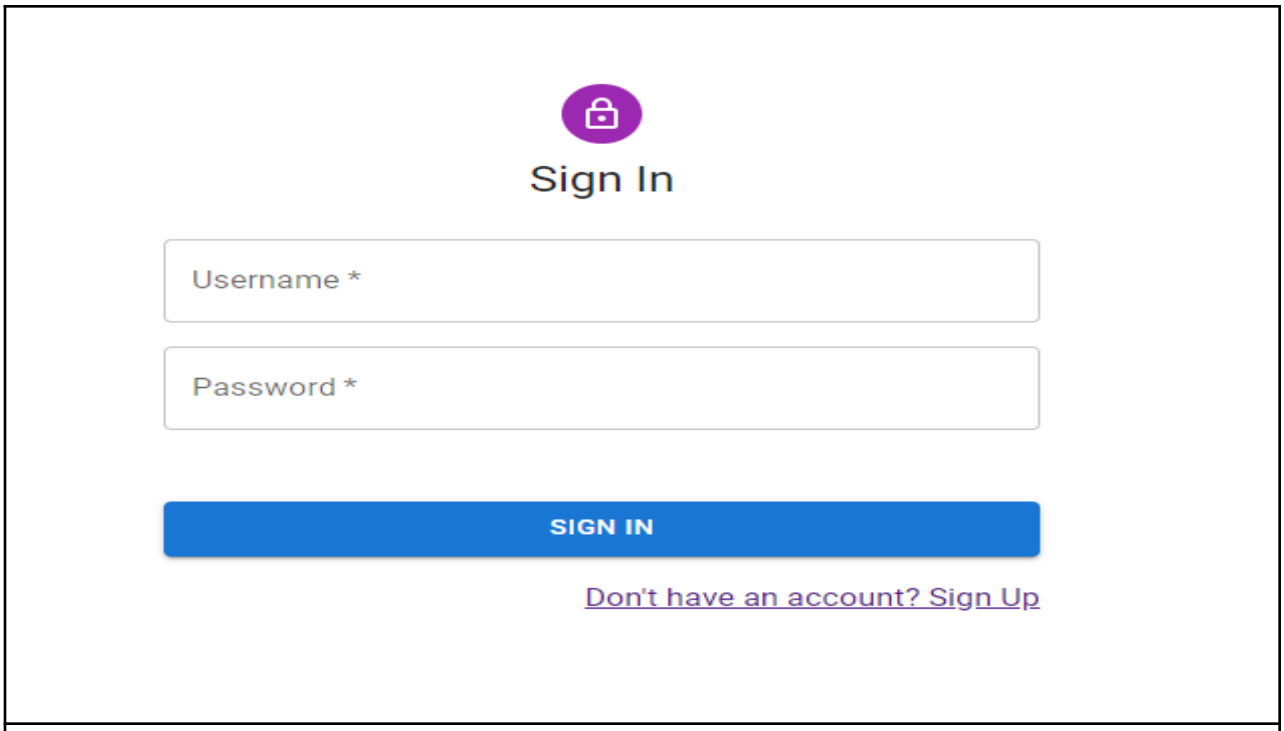


Figure 1. Login Page

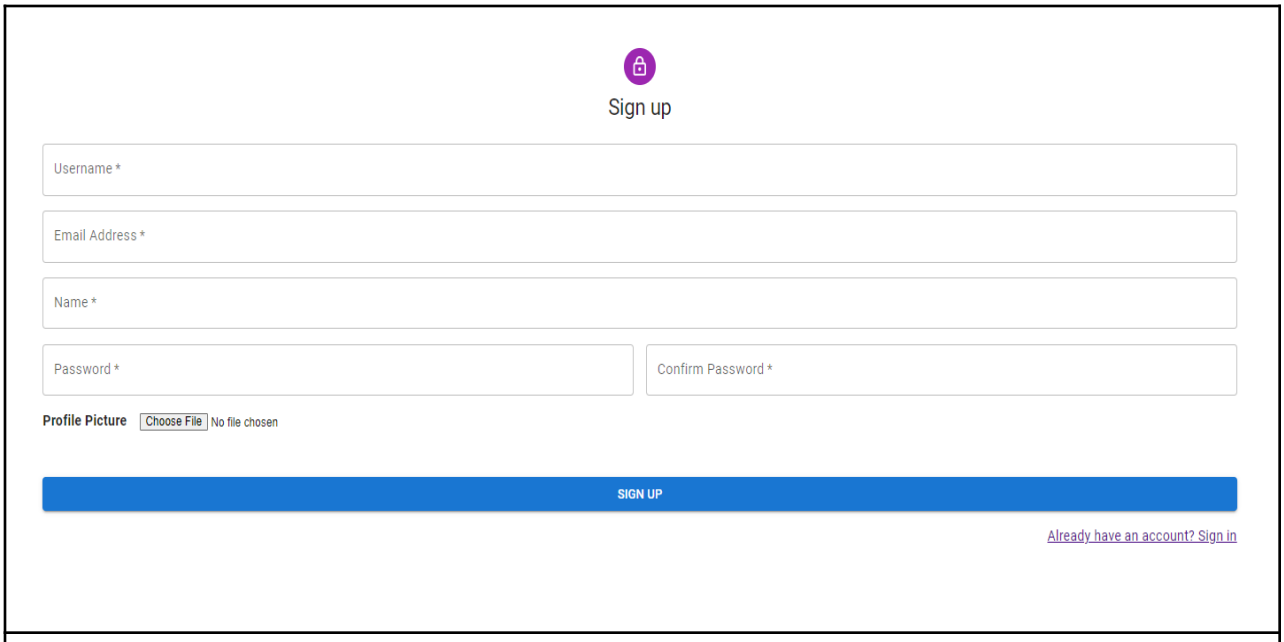


Figure 2. Register Page

Technologies Used:

There were a lot of technologies and tools I used to create this web application. I used React JS as the frontend having components for everything and making a responsive and user-friendly website. Node JS was used as the backend and handled requests and managed the server-side. It communicated with the frontend and the MySQL database to make everything work together. MySQL was used as the database for this project being able to store information securely through it.

Looking at some specific libraries I used throughout the project were Mutler which was used to handle file uploads, it was used within Node JS making sure there was a smooth process of uploading images. Axios was also used to communicate between the frontend and backend by exchanging data making sure interactions happened smoothly.

JSON Web Token (JWT) was a crucial key to making sure user authentication worked properly. Tokens were created securely making sure only the correct users could access their personalized homepage. Cookies played a important role with them by saving the JWT tokens and storing them allowing users to remain authenticated across different pages.

Another library used for security was Bcrypt which allowed for passwords to be hashed and stored securely added another extra layer of protection within the website.

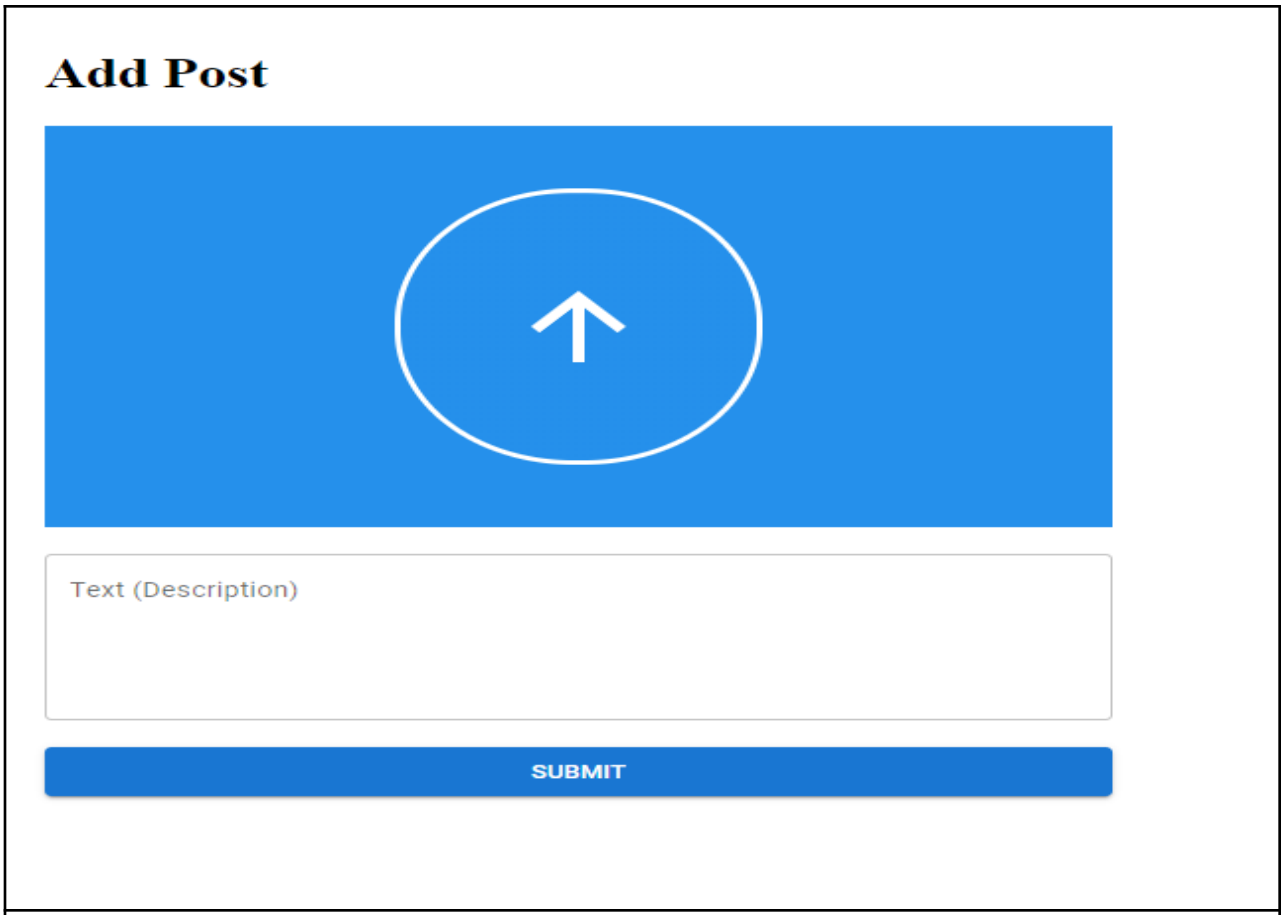


Figure 3. Add Post Page

Challenges:

There were a lot of challenges I faced throughout this project. One challenge that was particularly difficult was handling the file uploading with mutler. Trying to figure out how to integrate Mutler within the backend node.js side and Axios for the frontend was very confusing and involved a lot of troubleshooting and debugging.

Another challenge was working with complex logic to make sure the authentication worked properly with the JSON Web Tokens. Understanding how the token structure, encoding and verification process took some time to understand.

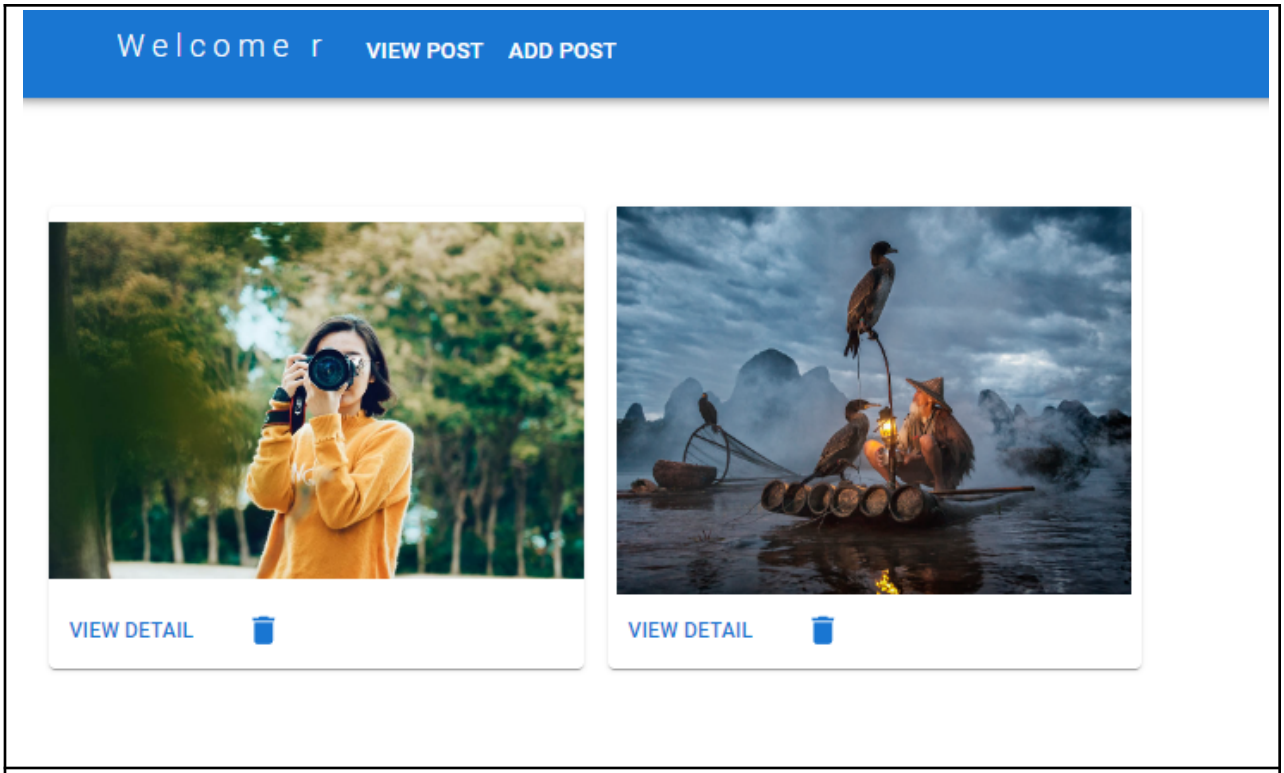


Figure 4. All Posts Page

Future Work:

This project still has more potential that I want to work on. The future work includes being able to view other users posts that you desire. Being able to comment, like, and see images your friends post. The idea is similar to Instagram but maybe more personal based and a simpler style.

What I learned from this project:

This project helped me significantly improve my knowledge of web development. All the challenges were a lesson showing me that you have to make the website seamless for users. Making the backend work with the frontend was an important thing I learned too since I have only ever done stuff with frontend. Also, working with the authentication gave me more knowledge on security within the website.

Conclusion:

Overall, this project was about having a website where users can express their feelings through a post and was able to achieve that. Using React as the frontend and Node as the backend with MySQL as the database and a multitude of libraries I was able to create this web application how I wanted it with good authentication and security yet a simple layout with Material UI. I want to expand this project in the future so users can interact with each other and that will be a challenge of its own.